CLOUD & DEVOPS

Kubernetes Workloads Visual Guide

Deployments, StatefulSets, DaemonSets, Jobs — pick the right controller for the job.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Controllers: Desired State for Pods

You declare replicas; the controller makes it so.

Deployment

stateless apps — web APIs

StatefulSet

stateful — DBs, queues

DaemonSet

one per node — agents

Job / CronJob

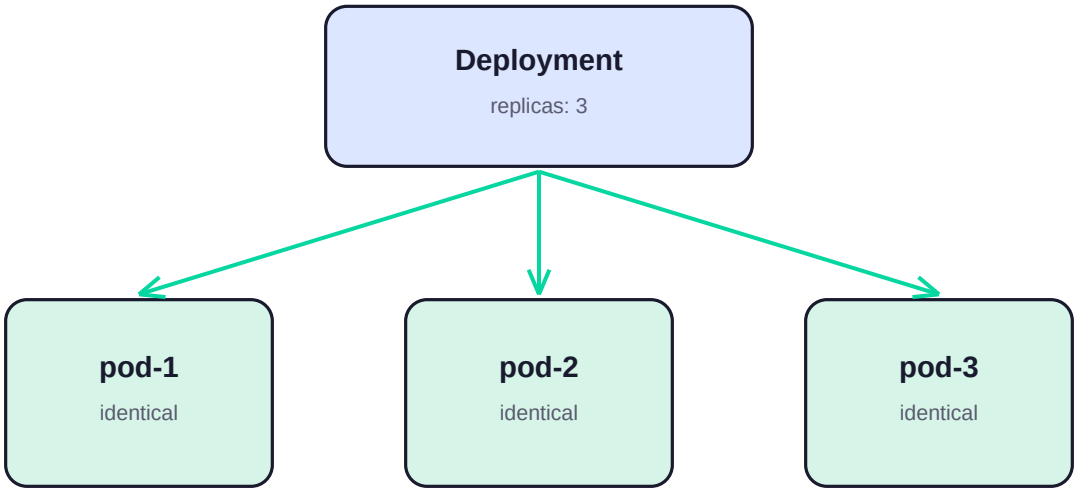
run to completion — batch

Pods are mortal — never create them directly. Controllers recreate, reschedule, and roll them.

STEP 2/6

Deployments: Stateless Workhorses

Identical, interchangeable pods behind a stable endpoint.



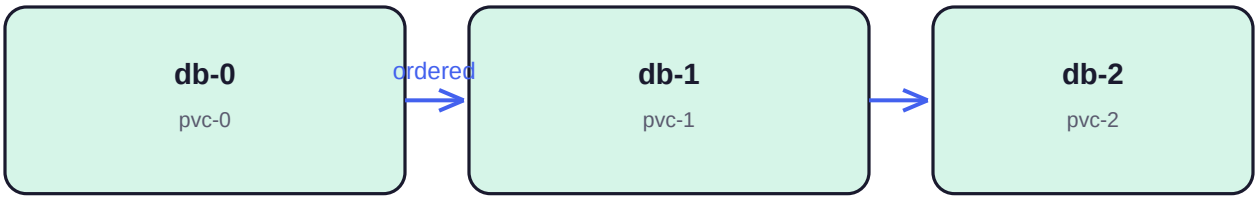
Pods are cattle: any pod can serve any request. State lives outside (DB, cache, queue).

- ReplicaSet (owned by the Deployment) keeps exactly N pods running.
- Rolling updates replace pods gradually — zero downtime by default.
- Rollbacks: `kubectl rollout undo` returns to the previous ReplicaSet.

STEP 3/6

StatefulSets: Identity Matters

Stable names, stable storage, ordered operations.



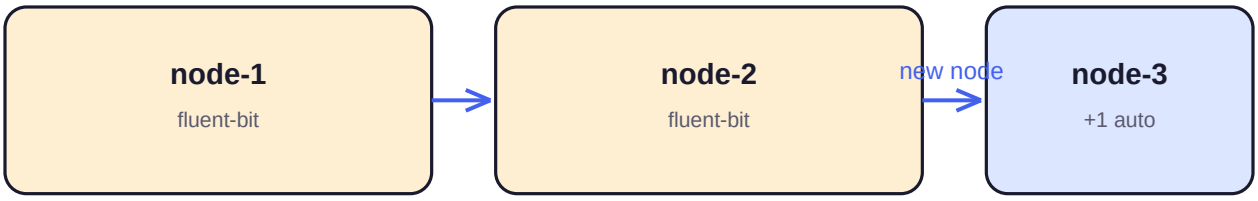
db-0 dies → a new db-0 reattaches pvc-0. Identity and data survive the pod.

- Use for: databases, Kafka/ZooKeeper, anything with quorum or local state.
- Headless Service gives each pod its own DNS: db-0.db, db-1.db.
- Ordered rolling updates: db-2, then db-1, then db-0 — never all at once.

STEP 4/6

DaemonSets: One Per Node

Cluster-wide agents that must run everywhere.



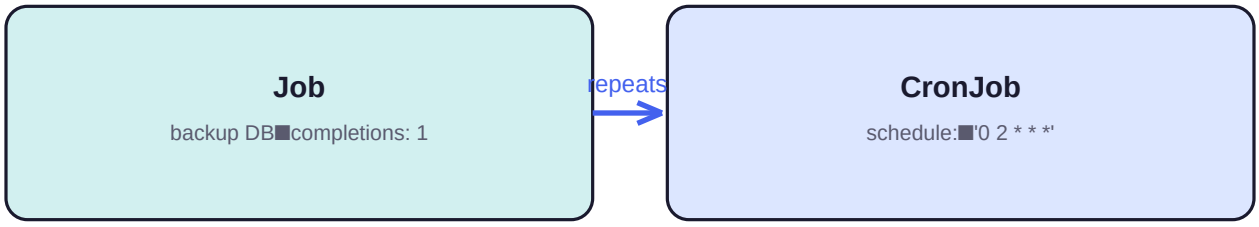
New node joins → DaemonSet schedules the pod automatically. No manual step.

- Classic uses: log shippers, monitoring agents, CNI plugins, storage drivers.
- Tolerations let DaemonSet pods land on tainted nodes (e.g. GPU nodes).
- Updates roll node by node — the agent fleet never goes fully dark.

STEP 5/6

Jobs & CronJobs: Run to Completion

Not 'keep running' — 'finish the work, then stop'.

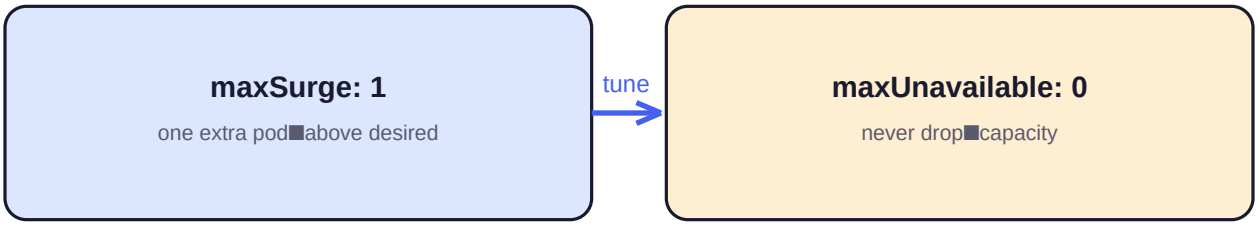


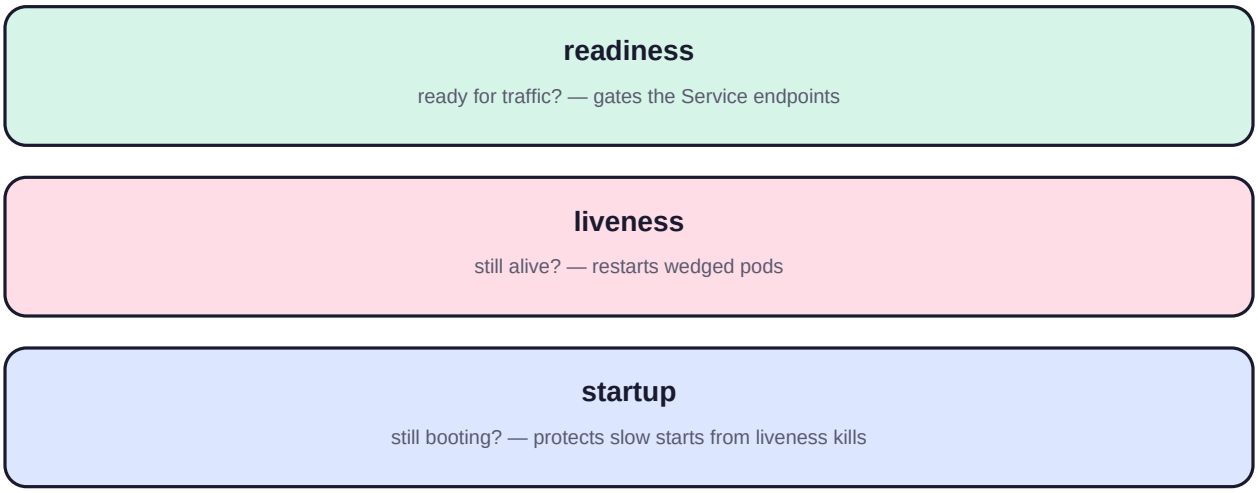
- Job: runs pods until N succeed; backoffLimit caps retries on failure.
- CronJob: cron schedule creates Jobs — reports, cleanup, batch ETL.
- Set activeDeadlineSeconds: a hung job must die, not run forever.
- History limits (successfulJobsHistoryLimit) keep etcd from filling with old jobs.

STEP 6/6

Rolling Updates & Probes

Update strategy + health probes = safe deploys.





STEP ✓/✓

Key Takeaways

Kubernetes Workloads — Remember

- Controllers maintain desired state — never create bare pods.
- Deployments for stateless; StatefulSets for stable identity + storage.
- DaemonSets put one agent pod per node, automatically on new nodes.
- Jobs run to completion; CronJobs schedule them — cap retries and deadlines.
- Rolling updates (surge/unavailable) + readiness/liveness/startup probes = safe deploys.

Keep exploring

Pair this guide with the matching video walkthroughs
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